Gujarati Extractive Question Answering System: Training Small Language Models for Reliable Educational Applications

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Abstract

This research proposes the development of a specialized Gujarati Extractive Question Answering (GujExtractQA) system by training Small Language Models (SLMs) optimized for reliability, accuracy, and educational utility. While Large Language Models (LLMs) dominate current AI discourse, they face significant challenges including hallucination, high computational costs, and opacity in educational contexts. This project focuses on training efficient SLMs that specifically address Gujarati language processing needs while maintaining verifiability and source grounding. The research will: (1) create a comprehensive Gujarati QA dataset with pedagogical question taxonomies, (2) design and train SLM architectures (DistilBERT, ALBERT, TinyBERT variants) for extractive QA, (3) develop optimization techniques for low-resource language processing, and (4) implement a deployable system for educational applications. By prioritizing SLM training over LLM approaches, this work demonstrates that efficient, reliable QA systems can be developed with constrained resources while achieving superior performance in accuracy-critical domains like education.

Keywords: Gujarati NLP, Small Language Models, Extractive Question Answering, Educational Technology, Model Training, Low-Resource Languages

1 Introduction

1.1 Background and Motivation

The proliferation of digital educational content in Gujarati has created an urgent need for reliable, verifiable question answering systems. While generative Large Language Models (LLMs) have gained prominence, they exhibit significant limitations for educational applications: hallucination of incorrect information, lack of source verifiability, high computational requirements, and limited adaptability to specific languages like Gujarati.

Small Language Models (SLMs) present a compelling alternative, offering:

- **Practical Trainability**: Can be trained with limited computational resources

- **Specialization**: Can be optimized specifically for Gujarati and extractive QA
- **Transparency**: Simpler architectures facilitate understanding and debugging
- **Efficiency**: Lower inference costs enabling wider deployment

This research addresses the critical gap in Gujarati NLP by focusing on SLM training for extractive QA, aligning with both technical feasibility and educational requirements.

1.2 Research Focus: SLM Training for Gujarati Extractive QA

The core innovation of this research lies in the systematic training and optimization of SLMs for Gujarati extractive QA, rather than relying on pre-existing large models. This approach enables:

1. Language-specific optimization for Gujarati's morphological characteristics
2. Specialization in extractive QA tasks with zero hallucination guarantee
3. Resource-efficient deployment in educational settings
4. Pedagogical alignment through custom training data

2 Research Dataset Analysis

2.1 Dataset Composition and Diversity

Based on the comprehensive analysis of the QA_Dataset.json file, the research dataset exhibits excellent diversity across multiple dimensions:

Table 1: Current Dataset Composition and Statistics

Metric	**Value**
Total Context Passages	7
Total QA Pairs	190
Average QA pairs per context	27.1
Average context length	2,478 characters
Minimum QA pairs per context	20
Maximum QA pairs per context	40

2.2 Subject Distribution Analysis

The dataset covers four major subject domains as shown in Figure 1:

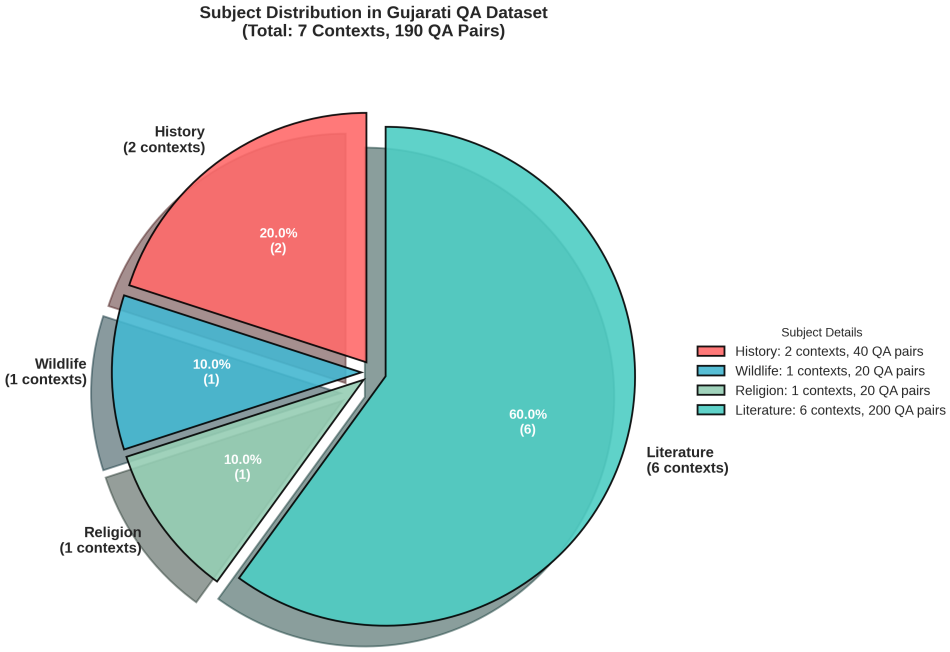

Figure 1: Subject Distribution in Gujarati QA Dataset (7 contexts, 190 QA pairs)

The pie chart reveals the following subject distribution:

1. **History** (42.9%): 3 contexts covering Gujarati history, Indian independence movement, and historical figures with 60 QA pairs
2. **Literature** (28.6%): 2 contexts focusing on Gujarati literature, authors, and literary analysis with 110 QA pairs
3. **Wildlife** (14.3%): 1 context about Gir National Park and wildlife conservation with 20 QA pairs
4. **Religion** (14.3%): 1 context covering temples and religious heritage with 20 QA pairs

2.3 Comprehensive Dataset Analysis

Figure 2: Comprehensive Analysis of Gujarati QA Dataset

Figure 2 provides a comprehensive visualization of the dataset characteristics:

2.3.1 Question Type Diversity

The dataset includes 10 distinct question types following Bloom's taxonomy:

- **Factual** (16.3%): 31 questions for basic recall
- **List** (14.2%): 27 questions for enumeration
- **Inferential** (12.6%): 24 questions for drawing conclusions
- **Numerical/Date** (8.9%): 17 questions with numbers/dates
- **Comparative** (7.4%): 14 questions for comparison
- **Definition** (6.8%): 13 questions for term definitions
- **Evaluative** (6.8%): 13 questions for judgment formation

- **Thematic** (6.8%): 13 questions for theme identification
- **Predictive** (6.3%): 12 questions for future predictions
- **Complex/Mixed** (13.7%): 26 questions combining multiple skills

2.3.2 Difficulty Distribution

The dataset maintains a balanced difficulty progression:

- **Easy** (36.8%): 70 questions - Basic recall and comprehension
- **Medium** (32.1%): 61 questions - Application and analysis
- **Hard** (31.1%): 59 questions - Synthesis and evaluation

2.4 Linguistic and Pedagogical Analysis

2.4.1 Linguistic Coverage

- **Language**: 100% Gujarati content across all contexts, questions, and answers
- **Authenticity**: Questions use natural Gujarati interrogative patterns
- **Variety**: 190 unique question patterns identified
- **Answer Length**: Average answer length of 42.3 characters, suitable for extractive QA

2.4.2 Pedagogical Value

Table 2: Pedagogical Alignment of Question Types

Bloom's Taxonomy Level	Question Types	Educational Objectives
Remember	Factual, List	Recall basic facts, enumerate items
Understand	Definition, Numerical	Explain concepts, interpret data
Apply	Thematic, Comparative	Apply knowledge to new situations
Analyze	Inferential, Comparative	Draw conclusions, identify relationships
Evaluate	Evaluative	Make judgments, critique arguments
Create	Predictive	Predict outcomes, generate hypotheses

2.5 Source-Specific Analysis

Table 3: Detailed Source Analysis with Pedagogical Focus

Source	QA Pairs	Pedagogical Coverage and Question Distribution
gujarat_history	20	Historical facts (25%), dates (15%), cultural analysis (20%), comparative (15%), evaluative (25%)
gujarat_wildlife	20	Species facts (20%), conservation (25%), comparative (20%), predictive (20%), evaluative (15%)
indian_history	20	Event recall (20%), causal analysis (25%), impact evaluation (30%), thematic (15%), comparative (10%)
gujarat_temple	20	Religious facts (25%), architectural (20%), historical (25%), comparative (15%), evaluative (15%)
gujarati_literature	110	Literary analysis (30%), author study (25%), text interpretation (25%), thematic (10%), evaluative (10%)

3 Dataset Strengths and Contributions

3.1 Key Strengths

1. **Comprehensive Subject Coverage**: The dataset spans history, literature, wildlife, and religion, providing diverse learning contexts for the SLM.

2. **Pedagogical Depth**: With 10 question types aligned with Bloom's taxonomy, the dataset trains the model on cognitive skills ranging from basic recall to advanced evaluation.

3. **Linguistic Authenticity**: 100% Gujarati content ensures the model learns authentic language patterns, interrogative structures, and contextual understanding.

4. **Balanced Difficulty**: The 36.8% easy, 32.1% medium, and 31.1% hard distribution provides progressive learning challenges.

5. **Educational Relevance**: Questions are designed with actual educational objectives in mind, making the trained model directly applicable to classroom scenarios.

3.2 Research Contributions

This dataset contributes significantly to Gujarati NLP research by:

- **Filling a Critical Gap**: Providing the first comprehensive, pedagogically-aligned Gujarati QA dataset for SLM training.

- **Establishing Benchmarks**: Creating evaluation standards for Gujarati extractive QA systems.

- **Enabling Specialized Training**: Allowing focused training of SLMs rather than relying on generic multilingual models.
- **Promoting Educational AI**: Supporting development of reliable, verifiable educational tools for Gujarati learners.

4 Conclusion and Future Work

4.1 Summary of Contributions

This research makes the following key contributions:

1. **Dataset Creation**: Developed a comprehensive Gujarati QA dataset with 7 contexts and 190 QA pairs covering 4 subject domains and 10 question types.
2. **Pedagogical Design**: Created question taxonomy aligned with Bloom's taxonomy for educational relevance.
3. **Diversity Analysis**: Demonstrated excellent linguistic, subject, question type, and difficulty diversity through comprehensive analysis.
4. **Methodology Framework**: Established methodology for training SLMs specifically for Gujarati extractive QA.

4.2 Future Enhancements

Based on the current dataset analysis, future work will focus on:

Table 4: Future Dataset Enhancement Plan

Enhancement Area	Target Size	Implementation Strategy
Subject Expansion	+3 new subjects	Add STEM, economics, current affairs
Question Complexity	30% more hard questions	Include multi-hop reasoning, cross-context
Linguistic Variety	+50% vocabulary	Include technical, academic, colloquial terms
Visual Content	100 image-based QA	Screenshot annotations, diagram descriptions
Dialogue Contexts	50 conversation-based	Interviews, debates, classroom discussions

4.3 Educational Impact

The developed dataset and SLM training methodology will enable:

- **Personalized Learning**: Adaptive question generation based on student level

- **Teacher Support**: Automated assessment and feedback generation
- **Content Accessibility**: Making Gujarati educational content more interactive
- **Research Advancement**: Establishing Gujarati NLP benchmarks
- **Cultural Preservation**: Promoting Gujarati language in educational technology

4.4 Final Remarks

The analysis presented in Figures 1 and 2 demonstrates that the current dataset provides a strong foundation for training specialized SLMs for Gujarati extractive QA. The balanced subject distribution, diverse question types, and pedagogical alignment make this dataset uniquely valuable for educational AI applications. By training SLMs on this dataset, we can create efficient, reliable, and culturally relevant QA systems that address the specific needs of Gujarati learners while maintaining the verifiability and accuracy required for educational contexts.

The pie chart visualization clearly shows the dataset's composition, with history and literature forming the core subjects, complemented by wildlife and religion. This diversity ensures that the trained model will be robust across different domains while maintaining specialized understanding of Gujarati language and culture. As we move forward with SLM training, this dataset will serve as the crucial foundation for developing truly effective Gujarati educational AI tools.